Synopsis

Behavioral Segmentation for Targeted Marketing

Problem Id: 10

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- **Problem Description:**

The main aim of this project is to **analyze customer purchasing behavior and group similar customers together**. Businesses collect large amounts of customer data, but simply having the data does not always help them understand their customers. Therefore, this project focuses on dividing customers into different groups based on their **Annual Spending** and **Purchase Frequency**.

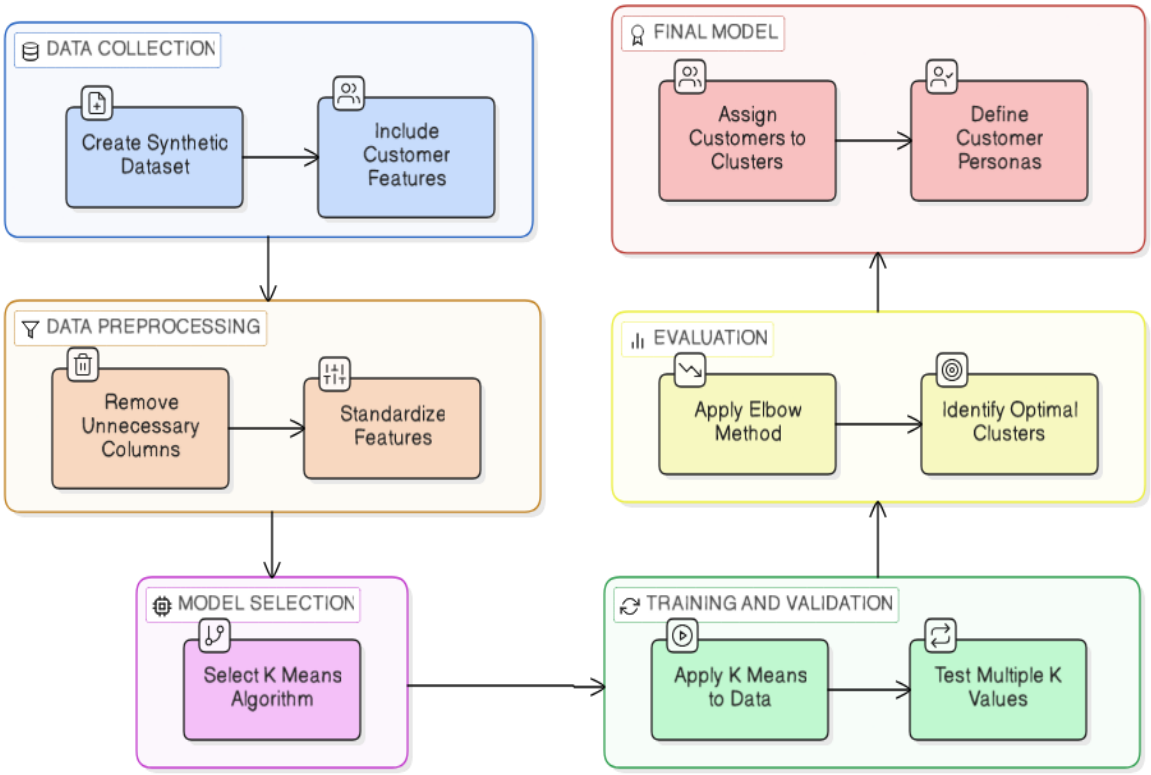
To achieve this, a **synthetic retail dataset** is created that represents customer purchasing patterns. The data is then preprocessed using **StandardScaler**, which helps bring all numerical values to a similar scale so that the analysis becomes fair and accurate. After preprocessing, the **K-Means clustering algorithm** is applied. This algorithm groups customers with similar spending habits into clusters.

To decide the correct number of clusters, the **Elbow Method** is used. It helps identify the point where increasing the number of clusters no longer significantly improves the grouping. Each resulting cluster represents a different **type of customer**, such as frequent buyers, high spenders, or low-engagement customers.

In real life, businesses use this type of analysis to understand their customers better. It helps companies create **targeted marketing strategies**, offer personalized promotions, improve customer satisfaction, and increase sales by focusing on the right group of customers.

- **Steps of Implementation:**

Fig. Architecture diagram



- **Expected Modules / Libraries to be Used**

1. pandas
   - Used for data manipulation and analysis. It helps in loading the dataset, handling missing values, selecting features, and organizing data in tabular format using DataFrames.

2. numpy
   - Provides support for numerical computations and array operations. It is useful for performing mathematical calculations and generating synthetic numerical data efficiently.

3. matplotlib
   - A visualization library used to create graphs and plots. In this project, it is mainly used to plot the **Elbow Method graph** and visualize clustering results.

4. seaborn
   - A statistical data visualization library built on top of matplotlib. It helps create more informative and visually appealing plots for exploratory data analysis.
5. scikit-learn (sklearn)
   - A machine learning library that provides tools for preprocessing and implementing machine learning algorithms.
6. sklearn.preprocessing.StandardScaler
   - Used to standardize numerical features so that they have a mean of 0 and a standard deviation of 1. This ensures that all features contribute equally during clustering.
7. sklearn.cluster.KMeans
   - Implements the **K-Means clustering algorithm** used to group customers into different segments based on similarities in their purchasing behavior.

- **<u>Expected Evaluation metrics and Plots to be used:</u>**

1. Elbow Method Plot

The Elbow Method will be used to determine the optimal number of clusters (K) for the K-Means algorithm. It plots the within-cluster sum of squared distances (inertia) against different values of K. The point where the graph forms an "elbow" indicates the most suitable number of clusters.

2. Scatter Plot for Cluster Visualization

A scatter plot will be used to visualize the clusters formed by the K-Means algorithm. The x-axis will represent Annual Spending, and the y-axis will represent Purchase Frequency. Different colors will represent different customer clusters, helping to clearly observe customer segments.

3. Cluster Distribution Plot

A simple plot will be used to show the number of customers in each cluster. This helps in understanding how customers are distributed among different groups.

- **<u>Required topics from the subject:</u>**

1. Strings

Strings are used to store and manipulate textual data such as customer names, cities, and categorical information in the dataset. They help in handling labels and performing basic text operations during data cleaning and preprocessing.

2. Lists and Arrays

Lists and arrays are used to store collections of values such as customer spending and purchase frequency. They allow efficient data handling and are commonly used when performing numerical operations and preparing data for machine learning models.

3. Functions

Functions help organize the code into reusable blocks. In this project, functions can be used for tasks such as data preprocessing, scaling the dataset, and applying clustering algorithms.

4. File Handling

File handling is used to read and write datasets. In this project, the dataset is loaded from a **CSV file**, processed for analysis, and the results can also be saved back into a file.

5. Libraries and Modules

Python libraries such as **pandas, numpy, matplotlib, and sklearn** are used as modules to perform data analysis, preprocessing, visualization, and machine learning operations efficiently.

- **Platform Required:**

Jupyter Notebook is used to run the machine learning pipeline in an organized manner. Each stage of the process such as **loading the dataset, preprocessing the data, applying StandardScaler, running the K-Means algorithm, and visualizing results using plots** can be executed in separate cells.

- **Books and Link Sources:**
  - Dataset Link : Click

1. Artificial Intelligence: A Modern Approach – Stuart Russell & Peter Norvig
2. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow – Aurélien Géron
3. Python for Data Analysis – Wes McKinney
4. Introduction to Machine Learning with Python – Andreas C. Müller & Sarah Guido

- https://scikit-learn.org
- https://pandas.pydata.org
- https://matplotlib.org
- https://scholar.google.com